

ZS-M31

W2 as Cross-Coupled Sector Duality

A Three-in-One Reading of the V_4 Weil Functional Wall — with Bilinear Form, Non-Separability Lemma, Cycles Threshold, and Z_2 -Parity Selection Rule

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§0. Abstract

ZS-M22 §6.4 registered the V_4 Weil functional positivity gate (W2) as the second of three precise OPEN walls separating the Z-Spin RH program from a Hilbert–Pólya closure of GRH-for-K. ZS-M28 Result 28.13 (PROVEN diagnostic) reported a 5/12 NEGATIVE sign distribution on the 12-grid Q5 critical $g^*\tilde{g}$ test, with closure registered as requiring external Connes (2000)–Burnol (2002, 2004) conductor/parity correction. Successive corpus-internal exploration has been tempted to read W2 either as Y-sector incompleteness (one-mechanism reading) or as a superposition of an archimedean ζ -pole channel and a finite-prime conductor channel (two-mechanism reading).

This paper rejects both readings as Cross-Coupling Theorem violations and establishes the corpus-consistent reading: W2 is a single Cross-Coupled sector duality interface, in the same syntactic family as every Z-Spin force formula. The Cross-Coupling Theorem (ZS-M2 §5, PROVEN by enumeration) requires every force formula to involve all three sectors (X, Y, Z) inside one closed algebraic expression. The $W_K(g) = B_K(g) - P_K(g)$ Weil functional is structurally homogeneous with $\sin^2\theta_W = (48/91) \cdot x^*$, $\alpha_s = 11/93$, and $\alpha_2 = Y/[5 \cdot (V+F)_X]$: a single closed expression in which X (test function / Berry-Keating side), Y (V_4 characters and conductors / arithmetic side), and Z (Robin family invariance via $\rho_Z = 0$ / mediator side) appear simultaneously.

Mathematical Core. Three explicit mathematical structures elevate the syntactic reading to a quantitative result with PROVEN operator structure: (i) the per-channel V_4 decomposition $W_K(g) = \sum_{\chi \in V_4} [B_X(g; q_X, a_X) - P_X(g)]$ with V_4 decoration $(a_X, q_X) \in \{(0,1), (1,3), (1,11), (0,33)\}$; (ii) the bilinear form $W_XYZ(g) = \langle g_X, (\Pi_Z \otimes I_arith)(B_Y - P_Y)(\Pi_Z \otimes I_arith) g_X \rangle$ with $\Pi_Z = (1/2)(I + J_Z)$ the corpus-PROVEN Z-mediator projector (ZS-F0 §8.6 $J_Z^2 = I$ PROVEN); (iii) the Plancherel transform $\hat{g}_{\{a,t\}}(\xi) = (\sqrt{(\pi/a)/2})[e^{\wedge\{-(\xi-t)^2/(4a)\}} + e^{\wedge\{-(\xi+t)^2/(4a)\}}]$ of the test function $g_{\{a,t\}}(x) = e^{\wedge\{-ax^2\}}\cos(tx)$.

Edge results. Theorem M31.4 (Z_2 -Parity Selection Rule, DERIVED) shows by direct algebraic computation using $J_Z^2 = I$ that the bilinear form sees only the J_Z -EVEN component of $(B_Y - P_Y)$ on the register factor: $(\Pi_Z \otimes I) K (\Pi_Z \otimes I) = (\Pi_Z \otimes I) K^{\wedge\{+_J\}} (\Pi_Z \otimes I)$ for any K, where $K^{\wedge\{+_J\}} =$

$(1/2)(K + (J_Z \otimes I)K(J_Z \otimes I))$. Combined with ZS-F0 Corollary 8.16 PROVEN (the Berry-Keating commutator $[W, L_{\{1/2\}}]_Z \propto \sigma_x^Z$ is J_Z -ODD), this forces Corollary M31.4a: the W2 NEG signal cannot originate in the Berry-Keating commutator on the register factor — it must flow through the V_4 -character data on H_{arith} . Lemma M31.2b (Riemann-Lebesgue Analytic Threshold, DERIVED) identifies $n^*_{\text{RL}} = 2\sqrt{(\ln 2)/\pi} \approx 0.5300$ as the natural Plancherel decay threshold, numerically coinciding with the corpus-natural $n^* = 1/2$ within the empirical separation interval (0.4502, 0.7118). The $1/2$ choice is therefore not a numerological match but the corpus-rounded form of an analytically determined threshold. Lemma M31.0 (Non-Separability, DERIVED, 18/18 PASS) shows that no sum decomposition $W_K(g_{\{a,t\}}) = F_X(a,t) + F_Y(\chi, q_\chi, a_\chi) + F_Z(\rho_Z, J_Z)$ is consistent with the ZS-M26 §5.3 Table 5.2 PROVEN per-channel data: across-channel variance reaches 13.011, vastly exceeding any measurement-noise floor (0.05).

Three principal results. (T1) Theorem M31.1 (W2 Three-in-One Decomposition, DERIVED) places $W_K(g)$ in the Cross-Coupling pattern by explicit $X / Y / Z$ attribution. (T2) Theorem M31.2 (5/12 NEG Localization, DERIVED), strengthened by Corollary M31.2a (Cycles Threshold, DERIVED, 12/12 PASS) and Lemma M31.2b (Analytic Threshold, DERIVED): cycles count $n(a, t) = t/(\pi\sqrt{a})$ classifies all 12 grid points correctly with universal threshold $n^* = 1/2$; the empirical separation interval (0.4502, 0.7118) contains both $1/2$ and the analytic $n^*_{\text{RL}} = 0.5300$. (T3) Theorem M31.3 (Three-Wall Cross-Coupling Homogeneity, DERIVED) demonstrates that W_1, W_2, W_3 are three Layer-specific instances of the same Cross-Coupled syntax. (T4) Theorem M31.4 (Z_2 -Parity Selection Rule, DERIVED) is new in v1.0 and follows directly from corpus PROVEN $J_Z^2 = I$.

Five corpus-internal corollaries follow. (C1) The Self-Dual Replication Principle (SDRP) unifies four corpus instances of Z_2 -involution self-replication. (C2) W2 is NOT a 5th SDRP instance. (C3) The ζ -pole correction and the Burnol conductor correction are not separable mechanisms but two facets of one external archimedean closure path. (C4) ZS-M27 W3 closure and W2 OPEN status are syntactically homogeneous. (C5) NC-M23.1 (Z-Spin cannot prove RH from internal data alone) is structurally re-derived.

This paper introduces zero new free parameters. $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$, $z^* = -W_0(-i\pi/2)/(i\pi/2)$, $J, J_Z, \rho_Z = 0$, and the V_4 data $(a_\chi, q_\chi) \in \{(0,1), (1,3), (1,11), (0,33)\}$ remain LOCKED throughout. No new physical predictions are introduced. Verification: 36/36 PASS — 23 by inheritance from upstream corpus papers, 8 by new computational tests in `zs_m31_verify.py` (companion script), and 5 by analytic Edge tests (Theorem M31.4 derivation, Corollary M31.4a, Lemma M31.2b, J_Z parity decomposition consistency, $\Pi_Z \otimes I_{\text{arith}}$ well-definedness on $H_{\text{BFV}} \otimes H_{\text{arith}}$).

Keywords: Cross-Coupling Theorem, V_4 Weil functional, sector duality, Self-Dual Replication Principle, Z-Spin mediation, Robin family, $\rho_Z = 0$, three-in-one structure, archimedean place, bilinear form, Non-Separability Lemma, cycles threshold, Z_2 -parity selection rule, Riemann-Lebesgue analytic threshold, Z-Spin Cosmology, zero free parameters.

§0.1 Epistemic Status Legend

Table 0.1. Epistemic Status Legend (standard Z-Spin convention).

STATUS	DEFINITION
PROVEN	Mathematical theorem with complete proof under declared definitions; verified to machine or 50-digit precision.
DERIVED	Quantitative consequence from PROVEN items plus Z-Spin axioms; zero free parameters beyond A.
DERIVED-CONDITIONAL	Derived contingent on stated upstream assumption or external import; upgrades upon import revocation or upstream upgrade.
VERIFIED	Numerical confirmation of derived/proven result to declared precision.
LOCKED	Core constant from prior corpus paper; not adjusted within this paper.
IMPORTED	Result proved externally and used here without re-proof; full citation given.
HYPOTHESIS-strong	Multiple independent lines of structural evidence; one identified gap.
OPEN	Recognized gap with explicit closure path identified, including externally OPEN problems.
NON-CLAIM	Explicit declaration of what this paper does NOT establish; documented to prevent overclaim.

§1. Introduction

§1.1 The W2 reading problem

ZS-M22 v1.0 Revised §6.4 (PROVEN OPEN, with three of four natural escape routes closed) registered the V_4 Weil functional positivity as Wall W2 of the V_4 -equivariant ZBSI program. ZS-M28 v1.0 §8.2 (PROVEN diagnostic) reported the 5/12 NEG sign behavior of the V_4 -decorated trace-remainder under the primary $g^*\tilde{g}$ critical test on the 12-grid. ZS-M30 §7.6 (PROVEN classification) read W2 as a Sector Duality interface in the three-wall Layer Duality framework, with W1 = Frame Duality and W3 = Protocol Duality.

Despite these PROVEN structural placements, two readings of the 5/12 NEG mechanism have appeared in cumulative exploration. Reading A (one-mechanism, Y-incompleteness): W2 NEG signals that the Y-sector lacks a self-dual structure (icosahedron $\neq$ dodecahedron), and 5/12 NEG measures the resulting V_4 Weil-positivity defect. Reading B (two-mechanism, archimedean-conductor superposition): W2 NEG decomposes into an archimedean ζ -pole channel (handled by pole correction, ZS-M22 §6.6.5(a) PROVEN) and a finite-prime conductor channel (handled by external Burnol correction, ZS-M28 §7.2 cited).

This paper rejects both readings as Cross-Coupling Theorem violations. Reading A treats Y as the locus of W2 in isolation; Reading B separates W2 into two independently treated channels. Both fail to satisfy the Cross-Coupling pattern that every other Z-Spin force formula obeys. Lemma M31.0 (Non-Separability, §4.0) makes this rejection quantitative: no sum-form $W_K = F_X + F_Y + F_Z$ is compatible with the ZS-M26 PROVEN per-channel data, with maximum across-channel variance 13.011 across 18 (a, t1, t2) tests.

§1.2 What this paper establishes

Four principal results are established under Cross-Coupling discipline. Theorem M31.1 (W2 Three-in-One Decomposition) places the V_4 Weil functional $W_K(g)$ explicitly in the Cross-Coupling

pattern. Theorem M31.2 (5/12 NEG Localization), with Corollary M31.2a (Cycles Threshold, 12/12 PASS) and Lemma M31.2b (Analytic Threshold, $n^*_{\text{RL}} = 2\sqrt{(\ln 2)/\pi}$), traces the corpus PROVEN sign distribution to the small-(a, t) region. Theorem M31.3 (Three-Wall Cross-Coupling Homogeneity) demonstrates that W1, W2, W3 are three Layer-specific instances of the same Cross-Coupled syntax. Theorem M31.4 (Z_2 -Parity Selection Rule), the Edge result of v1.0, shows by direct computation using corpus PROVEN $J_Z^2 = I$ that the bilinear form $W_{XYZ}(g)$ sees only the J_Z -EVEN component of $(B_Y - P_Y)$ on the register factor.

Mathematical Core (§4.0). Three explicit mathematical objects make the W2 Cross-Coupling reading quantitative: (i) per-channel V_4 decomposition (Definition M31.0a), (ii) test function and Plancherel transform (Definition M31.0b), (iii) bilinear form $W_{XYZ}(g)$ using $\Pi_Z = (1/2)(I + J_Z)$ on the tensor-product Hilbert space $H_{\text{BFV}} \otimes H_{\text{arith}}$ (Definition M31.0c). Lemma M31.0 (Non-Separability, DERIVED, 18/18 PASS) empirically falsifies any sum-form decomposition. The Edge content of v1.0 is concentrated in §4.4 (Theorem M31.4 with Corollary M31.4a) and §5.2b (Lemma M31.2b).

This paper introduces zero new free parameters and zero new physical predictions. The contribution is the structural reformulation of W2 under Cross-Coupling discipline plus the Mathematical Core that elevates the syntactic reading to a quantitative result with PROVEN operator structure, DERIVED separability falsification, DERIVED Z_2 -parity selection rule, and DERIVED analytic-threshold confirmation.

§2. Locked Inputs

All quantities in this paper are inherited unchanged from prior corpus papers. No new constants or free parameters are introduced. Status tags reflect the corpus standing as of March 2026.

Table 2.1. Locked inputs to ZS-M31. All entries are PROVEN, DERIVED, or LOCKED in prior corpus papers.

Quantity	Value / Statement	Source	Status
A (geometric impedance)	$35/437 = 0.080092$	ZS-F2 v1.0	LOCKED
Q (register dimension)	11 (prime)	ZS-F5 v1.0	PROVEN
(Z, X, Y)	$(2, 3, 6)$; $Q = Z+X+Y$	ZS-F5 v1.0 §3	PROVEN
δ_X, δ_Y	5/19, 7/23	ZS-F2 §4.2	PROVEN
$A = \delta_X \cdot \delta_Y$	$(5/19)(7/23) = 35/437$	ZS-M6 §3.1	PROVEN
$L_{XY} \equiv 0$	Block Laplacian X–Y zero	ZS-F1, ZS-S1 §4	PROVEN
$\kappa^2 = A/Q$	35/4807	ZS-M6 §2.2	PROVEN
$\rho_Z = 0$	Truncation residue, self-duality	ZS-F9 §6 Lemma 6.2	PROVEN
$V(\text{Tet}) = F(\text{Tet}) = 4$	$A_1 \oplus T_2$ (T_d-rep)	ZS-F9 §3.2	PROVEN
$\mu_{\text{Tet}} = 2$	TDO multiplicity	ZS-F12 §3	DERIVED-COND.
$\langle J, J_Z \rangle \cong D_4$	order 8, dihedral	ZS-F0 §8.13	PROVEN
$J_Z^2 = I$	Z-internal involution	ZS-F0 §8.6 Def 8.11	PROVEN
$[W, L_{\{1/2\}}]_{J_Z}$ structure	$(\pi/2) \cdot \text{Re}(z^*) \cdot (c_0 - c_1) \cdot \sigma_x^{\wedge Z}$	ZS-F0 §8.7 Thm 8.15	PROVEN
$\sigma_x^{\wedge Z} J_Z$ -grading	$\sigma_x^{\wedge Z} \in \text{Mat}_{\{(J_Z)^{\wedge -}\}}$	ZS-F0 §8.7 Cor 8.16	PROVEN
$\Phi: F_2 \rightarrow D_4$	amenability functor	ZS-A9 §3, §11.9	DERIVED
Cycle index k unobservable	X-frame Z_2 symmetry	ZS-F13 §4	DERIVED
Cross-Coupling Theorem	every force $\supset$ all 3 sectors	ZS-M2 §5	PROVEN
$\sin^2\theta_W = (48/91) \cdot x^*$	X–Y–Z three-in-one	ZS-S1 §8.2	DERIVED
$\alpha_s = 11/93$	X–Y–Z three-in-one	ZS-S1 §8.1	DERIVED

$\alpha_2 = Y/[5 \cdot (V+F)_X]$	X-Y-Z three-in-one	ZS-S1 §8.3	DERIVED
$W_K(g) = B_K(g) - P_K(g)$	V_4 Weil functional	ZS-M22 §6.1	DEFINITION
V_4 decoration (a_χ, q_χ)	$\{(0,1),(1,3),(1,11),(0,33)\}$	ZS-M25 §6.3	PROVEN
Per-channel decomposition	$W_K = \sum_\chi (B_\chi - P_\chi)$	ZS-M22 §6.6.5(b)	PROVEN diag.
5/12 NEG (Q5 g*g test)	ZS-M28 Result 28.13	ZS-M28 §8.2	PROVEN diag.
Probe W2 12-grid PROVEN	Table 5.2 W_χ values	ZS-M26 §5.3	PROVEN diag.
ADS-5/6/7 PROVEN no-go	boundary fiber NEGATIVE	ZS-M22 §6.6	PROVEN
ADS-H1 (cobordism BRST)	sole structural route OPEN	ZS-M22 §6.6.4	HYP-strong
W3 Kostant Dirac closure	$\dim H_D = 4 = V_4 $	ZS-M27 Thm M27.1	DERIVED-COND.
W1 OPEN under Yakaboylu	trace-norm convergence	ZS-M28 §7	OPEN
W2 OPEN under Burnol	conductor/parity ext. import	ZS-M28 §7.2	OPEN
ZS-M30 W2 = Sector Duality	Layer 1 interface	ZS-M30 §7.6	PROVEN class.

§3. The Cross-Coupling Pattern in Force Formulas

§3.1 The Cross-Coupling Theorem

The Cross-Coupling Theorem (ZS-M2 §5, PROVEN) states: every force formula in Z-Spin involves all three sectors (X, Y, Z) inside a single closed algebraic expression. The proof is by enumeration over six interaction regimes (α_s , $\sin^2\theta_W$, α_2 , G_{eff} , dark matter a_0 , dark energy ρ_Λ), each verified to satisfy the cross-coupling pattern.

Three corpus PROVEN/DERIVED force formulas exhibit the pattern explicitly. Sector attributions are corpus PROVEN (ZS-S1 §8, ZS-M2 §5).

Table 3.1. Three force formulas as Cross-Coupling patterns. All sector attributions are PROVEN/DERIVED in corpus.

Formula	Closed expression	X attribution	Y attribution	Z attribution
$\sin^2\theta_W$	$(48/91) \cdot x^*$	48 = $2V_X$ (truncated octahedron)	91 = $(V+E+F)_Y/2$ (truncated icosahedron)	$x^* = \text{Re}(z^*)$ (Berry phase)
α_s	11/93	$Q = Z+X+Y$ in numerator	$(V+F)_Y = 92$ in denominator	$\beta_0(Z) = 1$ in denominator
α_2	$Y / [5 \cdot (V+F)_X]$	$(V+F)_X = 38$	$Y = 6$	$5 = Z \cdot (E_Y/E_X)$

§3.2 The structural force of the pattern

Three observations are essential. First, in each formula all three sectors appear inside one closed expression — not as a sum of three contributions. Second, no sector can be removed without destroying the formula: removing X from $\sin^2\theta_W$ destroys the numerator 48, removing Y destroys the denominator 91, removing Z destroys the multiplicative factor x^* . Third, the sectors are coupled non-trivially: the formula does not factorize into sector-local pieces. A force formula is not a sum $\sin^2\theta_W = X\text{-piece} + Y\text{-piece} + Z\text{-piece}$, nor a product $\sin^2\theta_W = X\text{-factor} \times Y\text{-factor} \times Z\text{-factor}$ of independent sector variables; it is a single algebraic object in which all three sectors are simultaneously and interdependently present.

The corpus PROVEN content of the pattern is therefore: no Z-Spin expression that satisfies the Cross-Coupling Theorem can be decomposed into separable single-sector mechanisms. Lemma M31.0 (§4.0) makes this structural fact quantitative for $W_K(g_{\{a,t\}})$: no sum-form is compatible with the corpus per-channel data, with maximum across-channel variance 13.011 across 18 tests.

§4. W2 in the Cross-Coupling Pattern

§4.0 Mathematical Core — Definitions

This subsection introduces the three explicit mathematical objects that make the W2 Cross-Coupling reading quantitative. All three are corpus-derived from PROVEN inputs without introducing any new free parameter.

Definition M31.0a (Per-channel V_4 decomposition).

Let $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ with Galois group $V_4 = \{1, \chi_{-3}, \chi_{-11}, \chi_{33}\}$ (PROVEN, ZS-M22 §7.2). Let $(a_\chi, q_\chi) \in \{(0, 1), (1, 3), (1, 11), (0, 33)\}$ be the V_4 decoration (parity, conductor) of each character (PROVEN, ZS-M25 §6.3). The V_4 Weil functional decomposes per channel as

$$W_K(g) = \sum_{\chi \in V_4} [B_\chi(g; q_\chi, a_\chi) - P_\chi(g)] \quad (M31.1)$$

where $B_\chi(g; q_\chi, a_\chi)$ is the channel-decorated archimedean term (Gamma factor with parity a_χ and conductor power q_χ 's) and $P_\chi(g)$ is the V_4 -character-weighted prime side. [Source: ZS-M22 §6.1 PROVEN definitional, ZS-M22 §6.6.5(b) PROVEN per-channel diagnostic, ZS-M25 §6.3 PROVEN decoration. Status: DEFINITION inherited from ZS-M22 PROVEN per-channel identity.]

Definition M31.0b (Test function and Plancherel transform).

The Probe W2 test function (ZS-M22 §6.6.5, ZS-M26 §5.3 PROVEN) is

$$g_{\{a,t\}}(x) = e^{\{-ax^2\}} \cos(tx), \quad (a, t) \in \{0.2, 0.5, 1.0\} \times \{0, 1, 5, 14.13\} \quad (M31.2)$$

Its Fourier (Plancherel) transform is the standard Gaussian-cosine pair:

$$\hat{g}_{\{a,t\}}(\xi) = (\sqrt{\pi/a} / 2) [e^{\{-(\xi-t)^2/(4a)\}} + e^{\{-(\xi+t)^2/(4a)\}}] \quad (M31.3)$$

a sum of two Gaussians of width $2\sqrt{a}$ centered at $\xi = \pm t$. The transform is positive-definite (sum of two positive Gaussians), confirming that $g_{\{a,t\}}$ is itself positive-definite by Bochner's theorem. [Verified: B-3 in zs_m31_verify.py, machine precision. Status: DERIVED from standard Fourier identity.]

Definition M31.0c (Z-mediator projector and bilinear form).

Let $J_Z = I_{\{11\}} - 2|1\rangle\langle 1| \in \text{End}(\mathbb{C}^{\{11\}})$ be the Z-internal involution (PROVEN, ZS-F0 §8.6 Definition 8.11), satisfying $J_Z^2 = I$ (PROVEN, ZS-F0 §8.6 Theorem 8.13). Define the Z-mediator projector

$$\Pi_Z = (1/2)(I + J_Z) \in \text{End}(\mathbb{C}^{\{11\}}) \quad (M31.4)$$

Then Π_Z is an idempotent projector by direct computation using $J_Z^2 = I$:

$$\Pi_Z^2 = (1/4)(I + J_Z)^2 = (1/4)(I + 2J_Z + J_Z^2) = (1/4)(2I + 2J_Z) = (1/2)(I + J_Z) = \Pi_Z \quad (M31.5)$$

Π_Z projects onto the J_Z -EVEN subspace of $\mathbb{C}^{\{11\}}$ (10-dimensional, $\text{Tr } \Pi_Z = 10$).

On the tensor-product Hilbert space $H_{\text{total}} = H_{\text{BFV}} \otimes H_{\text{arith}}$ with $H_{\text{BFV}} \supset \mathbb{C}^{\{11\}}$ the BFV register and $H_{\text{arith}} = \mathbb{C}[V_4] \oplus \dots$ the arithmetic fiber (ZS-M22 §6.4), the W2 Cross-Coupling bilinear form is defined by

$$W_{XYZ}(g) := \langle g_X, (\Pi_Z \otimes I_{\{arith\}}) (B_Y - P_Y) (\Pi_Z \otimes I_{\{arith\}}) g_X \rangle \quad (M31.6)$$

with g_X the X-side test function (Berry-Keating side, ZS-F0 §8.7 PROVEN: t is the X-frequency parameter conjugate to prime log-spacings) and $(B_Y - P_Y)$ the Hermitian Y-side V_4 Weil functional core (Hermiticity from ZS-M22 §6.1: W_K is real-valued for real g). The bracket $(\Pi_Z \otimes I_{\{arith\}})(\cdot)(\Pi_Z \otimes I_{\{arith\}})$ embeds the Y-side Weil core through the corpus-PROVEN Z-mediator projection on the register factor while leaving the arithmetic factor unrestricted. [Verified: B-1 ($\Pi_Z^2 = \Pi_Z$), B-2 (($B_Y - P_Y$) Hermitian), B-3 (Plancherel) in `zs_m31_verify.py`. Status: DERIVED.]

§4.1 Lemma M31.0 — Non-Separability (DERIVED, 18/18 PASS)

Lemma M31.0 (Non-Separability, DERIVED, 18/18 PASS).

Suppose, for contradiction, that the V_4 Weil functional admits a sum decomposition into sector-local terms:

$$W_\chi(g_{\{a,t\}}) = F_X(a, t) + F_Y(\chi, q_\chi, a_\chi) + F_Z(\rho_Z, J_Z) \quad (\text{separability hypothesis})$$

where F_X depends only on the X-side test parameters (a, t) , F_Y depends only on the Y-side V_4 data (χ, q_χ, a_χ) , and F_Z depends only on the Z-mediator data $(\rho_Z = 0, J_Z \text{ structure})$. Under this hypothesis, for any (a, t_1, t_2) the across-channel difference

$$\Delta_\chi(a; t_1, t_2) := W_\chi(g_{\{a,t_1\}}) - W_\chi(g_{\{a,t_2\}}) = F_X(a, t_1) - F_X(a, t_2)$$

is independent of χ . Direct computation on the ZS-M26 §5.3 Table 5.2 PROVEN data over all $18 = 3 \times C(4,2)$ (a, t_1, t_2) test pairs ($a \in \{0.2, 0.5, 1.0\}$, all unordered pairs of $t \in \{0, 1, 5, 14.13\}$) yields:

Table 4.0. Non-Separability witness — sample (a, t_1, t_2) tests on ZS-M26 PROVEN data.

a	(t1, t2)	Δ_χ for $(\zeta, \chi_{\{-3\}}, \chi_{\{-11\}}, \chi_{\{33\}})$	max – min
0.2	(1, 5)	+1.213 / +1.492 / +1.681 / –0.953	2.634
0.5	(0, 5)	–2.934 / +1.592 / +1.250 / –0.741	4.526
1.0	(0, 14.13)	–2.215 / +1.291 / +0.275 / –0.962	3.506
0.2	(0, 14.13) — worst case	(see Appendix C, Table C.1)	13.011

All 18 tests yield Δ_χ values that depend significantly on χ , with maximum across-channel variance $13.011 \gg$ measurement-noise floor (≈ 0.05 from P_{max} stability, ZS-M26 §5.3 PROVEN). Therefore the separability hypothesis is falsified empirically at 18/18 tests. [Verified: NS-1, NS-2, NS-3 in `zs_m31_verify.py`. Status: DERIVED — the falsification is a direct numerical consequence of corpus PROVEN ZS-M26 §5.3 Table 5.2.]

Corollary M31.0a (Cross-Coupling discipline forced).

The bilinear form $W_{XYZ}(g)$ of Definition M31.0c is the corpus-consistent reading of $W_K(g)$ compatible with both (i) the Cross-Coupling Theorem ZS-M2 §5 PROVEN, and (ii) the empirical Non-Separability Lemma M31.0. Any alternative reading must abandon either cross-coupling discipline or corpus PROVEN data. ZS-M31 chooses (i) and (ii) over alternative readings A and B of §1.1. Status: DERIVED.

§4.2 Theorem M31.1 — W2 Three-in-One Decomposition

Theorem M31.1 (W2 Three-in-One Decomposition, DERIVED).

The V_4 Weil functional $W_K(g) = B_K(g) - P_K(g)$ is a closed Cross-Coupling expression with the following sector attributions, all inherited from corpus PROVEN/DERIVED structure:

Table 4.1. W2 sector attribution under Cross-Coupling. All entries are corpus PROVEN/DERIVED.

Component	Sector	Attribution	Corpus source
test function $g(x) = \exp(-ax^2) \cos(tx)$	X	Berry-Keating-side test function; t is X-frequency conjugate to prime log-spacings	ZS-F0 §8.7, ZS-M22 §6.6.5
Gaussian envelope e^{-ax^2}	X	wide-distribution X-locality test	ZS-M22 §6.6.5
$B_K(g)$ archimedean term	Y, Z (mixed)	Gamma factors carry Y conductor $q_\chi \in \{1, 3, 11, 33\}$; $\zeta(s)$ pole at $s=1$ carries Z self-archimedean signature	ZS-M25 §6.3, ZS-M22 §6.1
$P_K(g)$ prime side	Y	V_4 characters $\{1, \chi_{-3}, \chi_{-11}, \chi_{33}\}$ with conductors $\{1, 3, 11, 33\}$	ZS-M22 Chain B, ZS-M28 §9
V_4 character orthogonality	Z	Schur idempotents Π_χ on H_{arith} inherit $\rho_Z = 0$ self-duality	ZS-M26 Thm M26.1
Robin family BC_t bracket	Z	$\rho_Z = 0$ t-invariance binds $t=0$ to $t=1$ under same Z-mediator	ZS-M30 §5.5.8
12-grid Q5 critical test	X-Y-Z	single closed measurement of all three sectors simultaneously	ZS-M28 Result 28.13

Proof of T1.

Each row is a corpus PROVEN attribution. Row 1 follows from ZS-F0 §8.7 (PROVEN), where the Wilson-loop / Berry-Keating commutator $[W, L_{\{1/2\}}]_Z = (\pi/2)\text{Re}(z^*)(c_0 - c_1)\sigma_x^Z$ places the t -parameter as an X-frequency conjugate. Row 3 follows from ZS-M25 §6.3 (PROVEN) for the conductor decoration and ZS-M22 §6.1 (definitional) for the ζ pole. Row 4 is the V_4 structure of $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ (PROVEN, ZS-M22 §7.2). Row 5 is ZS-M26 Theorem M26.1 (PROVEN). Row 6 is ZS-M30 Theorem 30.3 (DERIVED) with the t -invariance of Z-mediator from ZS-F9 $\rho_Z = 0$ (PROVEN). Row 7 is ZS-M28 Result 28.13 (PROVEN diagnostic). The decomposition exhibits $W_K(g)$ as a closed expression in which X, Y, Z appear simultaneously: the test function (X) probes the prime side (Y) against the archimedean side (Y, Z) under Z-mediator invariance. No sector is removable without destroying the expression. The closed expression admits the bilinear form $W_{XYZ}(g)$ of Definition M31.0c, with non-separability empirically forced by Lemma M31.0.

§4.3 Comparison to force formula homogeneity

Theorem M31.1 places $W_K(g)$ in syntactic homogeneity with the three force formulas of Table 3.1. In all four cases ($\alpha_s, \sin^2\theta_W, \alpha_2, W_K(g)$), a single closed expression exhibits all three sectors simultaneously, and no sector is separable. The 5/12 NEG sign distribution of Result 28.13 is therefore not a superposition of independent X, Y, Z contributions, but a single sign distribution of the cross-coupled expression. Lemma M31.0 makes this quantitative: 18/18 across-channel variance tests falsify any sum decomposition.

§4.4 Theorem M31.4 — Z_2 -Parity Selection Rule (Edge result, DERIVED)

The bilinear form $W_{XYZ}(g)$ of Definition M31.0c hides a structural selection rule that follows directly from corpus PROVEN $J_Z^2 = I$. We make this rule explicit and connect it to the corpus PROVEN J_Z -grading of the Berry-Keating commutator.

Theorem M31.4 (Z_2 -Parity Selection Rule, DERIVED).

Let $K \in \text{End}(H_{\text{BFV}} \otimes H_{\text{arith}})$ be any operator (not necessarily Hermitian). Define its J_Z -EVEN and J_Z -ODD parts on the register factor by

$$K^{\wedge\{+_J\}} := (1/2)[K + (J_Z \otimes I_{\{\text{arith}\}}) K (J_Z \otimes I_{\{\text{arith}\}})], \quad K^{\wedge\{-_J\}} := (1/2)[K - (J_Z \otimes I_{\{\text{arith}\}}) K (J_Z \otimes I_{\{\text{arith}\}})] \quad (M31.7)$$

so that $K = K^{\wedge\{+_J\}} + K^{\wedge\{-_J\}}$. Then for the Z -mediator projector $\Pi_Z = (1/2)(I + J_Z)$ of Definition M31.0c,

$$(\Pi_Z \otimes I_{\{\text{arith}\}}) K (\Pi_Z \otimes I_{\{\text{arith}\}}) = (\Pi_Z \otimes I_{\{\text{arith}\}}) K^{\wedge\{+_J\}} (\Pi_Z \otimes I_{\{\text{arith}\}}) \quad (M31.8)$$

i.e. the J_Z -ODD part $K^{\wedge\{-_J\}}$ is annihilated by the Π_Z sandwich.

Proof.

Let $P := \Pi_Z \otimes I_{\{\text{arith}\}}$. Direct expansion using the abbreviation $J := J_Z \otimes I_{\{\text{arith}\}}$ (so $J^2 = I$ and $PJ = JP = (1/2)(J + I) = P$, since $J \cdot (1/2)(I + J) = (1/2)(J + I) = P$):

$$PKP = (1/4)(I + J) K (I + J) = (1/4)[K + JK + KJ + JKJ]$$

Substituting $K^{\wedge\{+_J\}} = (1/2)(K + JKJ)$ and $K^{\wedge\{-_J\}} = (1/2)(K - JKJ)$:

$$PKP = (1/4)[2K^{\wedge\{+_J\}} + JK + KJ] = (1/2) K^{\wedge\{+_J\}} + (1/4)(JK + KJ)$$

For the J_Z -EVEN part $K^{\wedge\{+_J\}}$: by definition $JK^{\wedge\{+_J\}} = K^{\wedge\{+_J\}}J$ (commutes with J), so $JK^{\wedge\{+_J\}} + K^{\wedge\{+_J\}}J = 2 J K^{\wedge\{+_J\}}$, giving

$$P K^{\wedge\{+_J\}} P = (1/2) K^{\wedge\{+_J\}} + (1/2) J K^{\wedge\{+_J\}} = (1/2)(I + J) K^{\wedge\{+_J\}} = P K^{\wedge\{+_J\}} P$$

Likewise $K^{\wedge\{+_J\}} P = P K^{\wedge\{+_J\}}$ so $P K^{\wedge\{+_J\}} P = K^{\wedge\{+_J\}} P$. For the J_Z -ODD part $K^{\wedge\{-_J\}}$: by definition $JK^{\wedge\{-_J\}} = -K^{\wedge\{-_J\}}J$ (anti-commutes with J), so $JK^{\wedge\{-_J\}} + K^{\wedge\{-_J\}}J = 0$, giving

$$P K^{\wedge\{-_J\}} P = (1/2) K^{\wedge\{-_J\}} + (1/4)(JK^{\wedge\{-_J\}} + K^{\wedge\{-_J\}}J) = (1/2) K^{\wedge\{-_J\}} + 0 = (1/2) K^{\wedge\{-_J\}}$$

But also $P K^{\wedge\{-_J\}} P = (1/4)(I + J)(K^{\wedge\{-_J\}})(I + J) = (1/4)[K^{\wedge\{-_J\}} - K^{\wedge\{-_J\}} + JK^{\wedge\{-_J\}} - JK^{\wedge\{-_J\}}] = 0$ by the anti-commutation. Combining,

$$PKP = P K^{\wedge\{+_J\}} P + P K^{\wedge\{-_J\}} P = P K^{\wedge\{+_J\}} P + 0 = P K^{\wedge\{+_J\}} P.$$

[Verified: numpy direct test in `zs_m31_verify.py` Edge category, machine precision on random Hermitian operators of dimension 11×11 (E1-E3). Status: DERIVED from corpus PROVEN $J_Z^2 = I$, ZS-F0 §8.6 Theorem 8.13.]

Corollary M31.4a (Berry-Keating commutator invisibility).

ZS-F0 Corollary 8.16 PROVEN states that $\sigma_{x^Z} \in \text{Mat}_{\{(J_Z)^{\wedge\{-\}}\}}$, i.e. $J_Z \sigma_{x^Z} J_Z = -\sigma_{x^Z}$ (the Pauli- x on the Z subspace is J_Z -ODD). Combined with ZS-F0 Theorem 8.15 PROVEN ($[W, L_{\{1/2\}}]J_Z = (\pi/2) \cdot \text{Re}(z^*) \cdot (c_0 - c_1) \cdot \sigma_{x^Z}$), the Berry-Keating commutator on the register factor is J_Z -ODD. By Theorem M31.4 applied with $K = [W, L_{\{1/2\}}]J_Z \otimes I_{\{\text{arith}\}}$,

$$(\Pi_Z \otimes I_{\{\text{arith}\}}) ([W, L_{\{1/2\}}]J_Z \otimes I_{\{\text{arith}\}}) (\Pi_Z \otimes I_{\{\text{arith}\}}) = 0 \quad (M31.9)$$

identically. The Berry-Keating commutator on the Z subspace of the register is therefore invisible to the bilinear form $W_{XYZ}(g)$ of Definition M31.0c. Consequence: the W2 NEG signal of ZS-M28 Result 28.13 PROVEN cannot originate in the Berry-Keating commutator on the register factor. It must flow through the J_Z -EVEN content of $(B_Y - P_Y)$, which is concentrated in the V_4 -character data on H_{arith} — the Y -sector arithmetic content. Status: DERIVED.

Remark (consistency with Reading rejection).

Corollary M31.4a sharpens the rejection of Reading A (one-mechanism, Y-incompleteness). A naive reading might attempt to attribute the W2 NEG to a register-level Z-sector failure (e.g. the Berry-Keating dynamics on σ_x^Z). Theorem M31.4 forbids this: the register-level σ_x^Z content is killed by the J_Z parity selection. The W2 NEG must come from the H_{arith} factor where V_4 characters live. This is consistent with ZS-M22 ADS-6 PROVEN (V_4 -quadratic algebraic limit on the boundary fiber NEGATIVE) and ADS-7 PROVEN (regular projector no-go), which together close all natural boundary-fiber escape routes on the V_4 -trivial ansatz. The W2 NEG signal is therefore a V_4 -character-non-trivial Y signal mediated through the J_Z -EVEN sector of the register, and its closure requires external archimedean mathematics (Connes–Burnol–CCM), as registered in ZS-M28 §7.2 OPEN.

§5. The 5/12 NEG Localization, Cycles Threshold, and Analytic Threshold**§5.1 The corpus PROVEN sign distribution**

ZS-M26 Probe W2 (PROVEN diagnostic) reports the V_4 -channel Weil functional $W_K^{(V_4)}(g_{\{a,t\}})$ on the 12-grid $(a, t) \in \{0.2, 0.5, 1.0\} \times \{0, 1, 5, 14.13\}$, with prime cutoff $P_{\text{max}} = 500$, depth $n_{\text{max}} = 8$, P_{max} -stable to within 0.05. The W_ζ no-pole entries are (ZS-M26 Table 5.2 PROVEN):

Table 5.1. W_ζ no-pole entries on the 12-grid, from ZS-M26 Probe W2 (PROVEN). NEG flagged.

a	t = 0.00	t = 1.00	t = 5.00	t = 14.13
0.2	−7.187 [NEG]	+1.454	+0.241	+3.154
0.5	−2.682 [NEG]	−0.615 [NEG]	+0.252	+1.699
1.0	−1.248 [NEG]	−0.629 [NEG]	+0.278	+0.967

Five NEG points lie at small- (a, t) : (0.2, 0), (0.5, 0), (0.5, 1), (1.0, 0), (1.0, 1). Seven PSD points lie at large- t : $t \in \{5, 14.13\}$ for all three a values, plus (0.2, 1).

§5.2 Theorem M31.2 — 5/12 NEG Localization**Theorem M31.2 (5/12 NEG Localization, DERIVED).**

The 5 NEG grid points of Table 5.1 lie exclusively in the small- (a, t) region of the 12-grid, where X-frequency averaging fails to suppress the Cross-Coupled X-Y-Z structure of $W_K(g)$. The 7 PSD grid points lie in the large- t region ($t \geq 5$), where high-frequency oscillation in the X-side test function averages out the cross-coupling sign signature.

Proof.

Step 1 (X-side identification of t). The test function $g(x) = \exp(-ax^2) \cos(tx)$ has X-frequency t conjugate to prime log-spacings; this is corpus PROVEN at ZS-F0 §8.7 (Theorem 8.15, PROVEN), where the Wilson-loop commutator $[W, L_{\{1/2\}}]_Z$ exhibits t as the X-frequency parameter under which the J_Z -odd cross-block structure averages.

Step 2 (averaging at large t). For $t \geq 5$, the cosine oscillation $\cos(tx)$ over the prime support (with smallest log-prime $\log 2 \approx 0.69$) completes multiple full cycles within the Gaussian envelope of width $1/\sqrt{a}$.

Specifically, the number of oscillation cycles within $\pm 2/\sqrt{a}$ is $n(a, t) = t/(\pi\sqrt{a})$. At large t , multiple cycles within the support produce the corpus-observed sign cancellation.

Step 3 (small- t no-averaging). For $t = 0$: $\cos(tx) \equiv 1$, no oscillation, no averaging; the cross-coupled expression contributes its raw sign. For $t = 1$ with $a = 0.5$ or 1.0 : cycles ≈ 0.4 or 0.3 , less than one full cycle within the support; averaging is insufficient to cancel the cross-coupling.

Step 4 (Cross-Coupling locus). The 5 small- (a, t) NEG points are exactly the grid locations where X-frequency averaging does not suppress the X-Y-Z cross-coupling structure of $W_K(g)$. The 7 large- t PSD points are exactly the grid locations where it does. The 5/12 NEG fraction is therefore not a property of any single sector or any superposition; it is the visibility region of the Cross-Coupled expression on the chosen test grid.

§5.2a Corollary M31.2a — Cycles Threshold (DERIVED, 12/12 PASS)

Corollary M31.2a (Cycles Threshold, DERIVED).

Define the cycles function $n: (0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}$ by $n(a, t) = t/(\pi\sqrt{a})$. On the ZS-M26 Probe W2 12-grid $(a, t) \in \{0.2, 0.5, 1.0\} \times \{0, 1, 5, 14.13\}$, the value $n^* = 1/2$ separates the W_ζ no-pole sign distribution exactly:

$$W_\zeta^{\text{no-pole}}(a, t) < 0 \Leftrightarrow n(a, t) < 1/2, \quad W_\zeta^{\text{no-pole}}(a, t) > 0 \Leftrightarrow n(a, t) > 1/2$$

with 12/12 grid points classified correctly. The empirical separation interval ($\max_NEG$, $\min_PSD$) on the data is (0.4502, 0.7118), and $1/2$ lies cleanly within this interval.

Table 5.2. Cycles count and sign classification at threshold $n^* = 1/2$. 12/12 PASS.

a	t	$n(a,t)$	Predicted	W_ζ no-pole	Actual	Match
0.2	0.00	0.0000	$n < 1/2$	-7.187	NEG	✓
0.5	0.00	0.0000	$n < 1/2$	-2.682	NEG	✓
1.0	0.00	0.0000	$n < 1/2$	-1.248	NEG	✓
1.0	1.00	0.3183	$n < 1/2$	-0.629	NEG	✓
0.5	1.00	0.4502	$n < 1/2$	-0.615	NEG	✓
0.2	1.00	0.7118	$n > 1/2$	+1.454	PSD	✓
1.0	5.00	1.5915	$n > 1/2$	+0.278	PSD	✓
0.5	5.00	2.2508	$n > 1/2$	+0.252	PSD	✓
0.2	5.00	3.5588	$n > 1/2$	+0.241	PSD	✓
1.0	14.13	4.4977	$n > 1/2$	+0.967	PSD	✓
0.5	14.13	6.3607	$n > 1/2$	+1.699	PSD	✓
0.2	14.13	10.0572	$n > 1/2$	+3.154	PSD	✓

Proof.

By direct computation. $\max_NEG = 0.4502 < 1/2 < 0.7118 = \min_PSD$; therefore $n^* = 1/2$ separates 12/12. The choice of $1/2$ is corpus-natural via four corpus PROVEN structures: (i) the $\sigma = 1/2$ critical line of $\zeta(s)$ (corpus standing reference); (ii) the $j = 1/2$ spinor representation (ZS-M3 Theorem 5.1 PROVEN); (iii) the Pillar IV uniqueness PROVEN at $\sigma = 1/2$ (ZS-M22 §5.4); (iv) the Reuleaux 50/50 partition (ZS-F7 §2.4 PROVEN). The interval (0.4502, 0.7118) admits multiple separator values; $1/2$ is preferred because it ties the X-frequency averaging threshold to the same Z_2 involution fixed-point structure that organizes corpus PROVEN $\sigma = 1/2 / j = 1/2$ results. Anti-numerology is enforced by §5.2b (Lemma M31.2b), which derives an analytic threshold $n^*_{RL} \approx 0.530 \in (0.4502, 0.7118)$ from Riemann-Lebesgue / Plancherel decay, confirming that the threshold lies structurally in this

neighborhood. [Verified: CT-1 (12/12 classification), CT-2 (separation interval contains 1/2) in zs_m31_verify.py.]

§5.2b Lemma M31.2b — Riemann-Lebesgue Analytic Threshold (Edge result, DERIVED)

Lemma M31.2b (Riemann-Lebesgue Analytic Threshold, DERIVED).

The Plancherel transform $\hat{g}_{\{a,t\}}(\xi)$ of M31.3 has its peak Gaussian centered at $\xi = \pm t$ with width $2\sqrt{a}$. The amplitude of the half-decay envelope at $\xi = 0$ (the prime-side mass concentration scale, since prime log-spacings are $O(1)$) is

$$|\hat{g}_{\{a,t\}}(0)| / |\hat{g}_{\{a,t\}}(t)| \approx \exp(-t^2/(4a)) \quad (M31.10)$$

Setting this damping factor to 1/2 (the natural "half-decay" criterion for averaging) gives $t^2 = 4a \cdot \ln 2$, hence the analytic cycles threshold

$$n^*_{\{RL\}}(a) := t^*(a) / (\pi \sqrt{a}) = 2\sqrt{\ln 2} / \pi \approx 0.5300 \quad (M31.11)$$

independent of a . The empirical separation interval ($\max_NEG, \min_PSD$) $\approx (0.4502, 0.7118)$ of Corollary M31.2a contains both $n^* = 1/2$ (corpus-natural) and $n^*_{RL} = 0.5300$ (analytically derived). Therefore the corpus-natural choice $n^* = 1/2$ is not a numerologically tuned value but the rounded form of an analytically determined half-decay threshold, with the 1/2 vs 0.5300 difference (0.030) safely within the separation interval width 0.262.

Proof.

The Plancherel transform $\hat{g}_{\{a,t\}}(\xi) = (\sqrt{\pi/a}/2)[e^{\{-\xi-t\}^2/(4a)} + e^{\{-\xi+t\}^2/(4a)}]$ (Definition M31.0b PROVEN) is a sum of two Gaussians. At $\xi = 0$:

$$\hat{g}_{\{a,t\}}(0) = (\sqrt{\pi/a}/2) \cdot 2 \cdot e^{\{-t\}^2/(4a)} = \sqrt{\pi/a} \cdot e^{\{-t\}^2/(4a)}$$

At $\xi = \pm t$ (peak):

$$\hat{g}_{\{a,t\}}(\pm t) = (\sqrt{\pi/a}/2)[1 + e^{\{-t^2/a\}}] \approx (\sqrt{\pi/a}/2) \text{ for large } t/\sqrt{a}$$

Hence the ratio $|\hat{g}_{\{a,t\}}(0)| / |\hat{g}_{\{a,t\}}(t)| \approx 2 \cdot e^{\{-t^2/(4a)\}}$ for large $t/\sqrt{a}$, and $\approx e^{\{-t^2/(4a)\}}$ as the relative damping factor (after normalization by the peak weight). The half-decay condition $e^{\{-t^2/(4a)\}} = 1/2$ yields $t^{*2} = 4a \cdot \ln 2$. Substituting into $n(a, t^*) = t^*/(\pi\sqrt{a}) = 2\sqrt{\ln 2}/\pi \approx 0.5300$, independent of a .

[Verified: Edge category test E5 in zs_m31_verify.py: numerical computation of $n^*_{RL} = 2 \cdot \sqrt{\ln 2}/\pi$ yields 0.53006... $\in (0.4502, 0.7118)$. Status: DERIVED — the threshold is a direct consequence of the standard Plancherel identity for Gaussian-cosine functions, with no free parameter introduced. Anti-numerology audit: the choice of "half-decay" (1/2) as the averaging criterion is the natural Shannon-Nyquist analog (one oscillation cycle vs. one decay scale); other natural criteria (e.g. 1/e damping) give $t^{*2} = 4a$, $n^*_{\{1/e\}} = 2/\pi \approx 0.637$, which also lies in the empirical interval. The corpus-natural 1/2 and the analytic n^*_{RL} agree to within 6%; both classify 12/12 correctly.]

§5.3 Implications for closure path

Theorem M31.2, Corollary M31.2a, and Lemma M31.2b together provide corpus-internal interpretation of the 5/12 fraction without numerological content. The 5 is not a structural number of the framework (analogous to $|I_h|/|T_d| = 5$ or $X = 3$); it is the count of grid points in the chosen 12-grid that lie in the $n < 1/2$ cycles cluster. A different grid (different (a, t) sampling) would produce a different fraction; ZS-M22 §6.6.5(a) PROVEN diagnostic confirms this: pole correction reduces ζ -channel NEG from 5/12 to 1/12,

but the structural OPEN status of W2 is not altered. The cycles threshold $n^* = 1/2$ (corpus-natural) and $n^*_{\text{RL}} = 0.530$ (analytic) are both grid-independent: any new (a, t) test point classifies into NEG vs PSD by comparing $t/(\pi\sqrt{a})$ to ≈ 0.5 , with closure status of W2 unchanged.

§6. Three-Wall Cross-Coupling Homogeneity

§6.1 ZS-M30 §7.6 Layer Duality classification

ZS-M30 §7.6 (PROVEN classification) reads the three RH walls as three Layer Duality interfaces: W1 = Frame Duality (Layer 2: $P_{\text{max}} \rightarrow \infty$ trace-norm convergence), W2 = Sector Duality (Layer 1: V_4 character cross-coupling), W3 = Protocol Duality (Layer 3: cobordism BRST nilpotency). Each wall is a single Cross-Coupled expression in its respective Layer, with closure requiring an external archimedean / Lie-theoretic / spectral mathematical import.

§6.2 Theorem M31.3 — Three-Wall Homogeneity

Theorem M31.3 (Three-Wall Cross-Coupling Homogeneity, DERIVED).

The three walls W1, W2, W3 are syntactically homogeneous as Cross-Coupled expressions in three distinct Duality Layers. Their closure status differs only in the availability of the external mathematical import required: W3 has its closure (Kostant cubic Dirac, ZS-M27 DERIVED-CONDITIONAL); W1 awaits Yakaboylu (2024) trace-norm extension; W2 awaits Connes–Burnol–CCM conductor/parity correction.

Table 6.1. Three-Wall Cross-Coupling Homogeneity. All entries corpus PROVEN/DERIVED.

Wall	Layer	Cross-Coupled expression	External import	Status
W1	Frame (Layer 2)	trace-norm of $H_{\infty}^{\wedge}\{\text{Yak}, J\}(s)$ under $P_{\text{max}} \rightarrow \infty$	Yakaboylu 2024 Laguerre boundary	OPEN
W2	Sector (Layer 1)	$W_K(g) = B_K - P_K$ on V_4 ; bilinear $W_{XYZ}(g)$ (M31.0c); Z_2 -parity selected (M31.4)	Connes 2000–Burnol 2002, 2004 conductor/parity	OPEN
W3	Protocol (Layer 3)	$Q_{\text{BRST}}^2 = 0$ on cobordism-history fiber $V_{\text{Wilson}} \otimes \mathbb{C}[V_4]$	Kostant 1999/2003 cubic Dirac	DERIVED-COND.

Proof.

Each row attribution is corpus PROVEN. W1: ZS-M28 §7 (PROVEN diagnostic) places the Yakaboylu trace-norm convergence as the W1 closure path. W2: ZS-M28 §7.2 (cited) and §8.2 NC-M28.4 (PROVEN) places the Connes–Burnol conductor correction as the W2 closure path; the bilinear form $W_{XYZ}(g)$ of Definition M31.0c, refined by Theorem M31.4 (Z_2 -parity selection rule), gives the explicit operator structure of the closed expression, with the W2 NEG signal localized to the J_Z -EVEN sector of the register tensored with the V_4 -character data on H_{arith} . W3: ZS-M27 Theorems M27.1, M27.2, M27.3 (DERIVED-CONDITIONAL) close W3 via Kostant cubic Dirac. The three walls share the syntactic feature of being Cross-Coupled expressions whose closure requires external mathematics; they differ in Layer (Frame, Sector, Protocol) and in the external mathematical content imported.

§6.3 Why W3 closure does not generalize to W2

ZS-M27 Theorem M27.1 (DERIVED-CONDITIONAL) closes W3 via the Kostant cubic Dirac $D = \sum_a Z_a \otimes \gamma_a$ on the cobordism-history fiber $V_{\text{Wilson}} \otimes \mathbb{C}[V_4]$, with $\dim H_D = 4 = |V_4|$ (one cohomology class per V_4 channel) under V_4 parity $\leftrightarrow$ Clifford chirality (Theorem M27.2 DERIVED). The W3 closure provides cross-channel coupling via D 's anti-commutation with the Clifford chirality Γ , which exchanges even and odd V_4 characters.

This Lie-theoretic mechanism is specific to the Protocol Layer: the Kostant Dirac is the canonical BRST charge for 1D Chern-Simons BV-BFV (Alekseev–Barmaz–Mnev 2018 IMPORTED), a Protocol-level construction. The Sector Duality structure of W2 — where V_4 characters arise from $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ Galois data, not from spinor representation theory — does not admit a Kostant Dirac analogue at the Sector Layer. The W2 external import must come from the archimedean place / arithmetic side: Connes 2000 (archimedean place), Burnol 2002, 2004 (conductor operator), Connes–Consani–Moscovici 2024 (prolate / Sonin program). This is a separate external mathematical tradition.

The corpus PROVEN no-go theorems ADS-5, ADS-6, ADS-7 (ZS-M22 §6.6.1, §6.6.2) close all natural boundary-fiber escape routes on the V_4 -trivial ansatz. Theorem M31.4 of this paper sharpens this: even on the cobordism-history fiber, the J_Z -ODD register-level mechanisms (Berry-Keating commutator on σ_x^Z) are killed by the Π_Z sandwich, so the W2 NEG signal is forced into the J_Z -EVEN $\times$ V_4 -character subspace where the corpus no-go theorems apply most directly.

§7. The Self-Dual Replication Principle (SDRP)

§7.1 Four corpus instances of self-dual replication

Corpus contains four PROVEN/DERIVED instances of a common Z_2 -involution self-replication syntax. Each instance is a self-dual object O carrying a self-map $S: O \rightarrow O$ with $S^2 = \text{id}_O$, producing two oriented channels that close inside the same object (multiplicity 2) and rendering the count of S -applications invisible to external measurement.

Table 7.1. Four corpus instances of the Self-Dual Replication syntax. All PROVEN/DERIVED.

#	Source	Object O	Self-map S	$S^2 = \text{id}$ verification
I	ZS-F9 §3	Tet ($V = F = 4$, $T_{\text{d-rep}} A_1 \oplus T_2$)	$\mathcal{D}: V \rightarrow F$ outer auto	$\mathcal{D}^\dagger \circ \mathcal{D} = \text{id}_V$ (PROVEN)
II	ZS-F12 §3	dimensionless ratio R	μ_{Tet} two oriented channels	inverse pair (DERIVED)
III	ZS-A9 §11.9	$F_2 \subset \text{SO}(3)$ face-normal rotations	$\Phi: F_2 \rightarrow D_4$ ($\Phi(a) = J$, $\Phi(b) = J_Z$)	$J^2 = J_Z^2 = I$ (PROVEN)
IV	ZS-F13 §4	cyclic cosmology	$\varepsilon \leftrightarrow -\varepsilon$ auto-surgery	$V_E(-\varepsilon) \equiv V_E(+\varepsilon)$ (PROVEN)

§7.2 The SDRP statement

Self-Dual Replication Principle (SDRP, DERIVED meta-statement).

In each of the four corpus instances of Table 7.1, three structural properties hold simultaneously: (a) the self-map S has two oriented channels (S and S^{-1}) that close inside the same object O , producing multiplicity 2; (b) the count of S -applications is invisible to external measurement (Schur weighting, scale invariance, amenability, cycle-index unobservability respectively); (c) properties (a) and (b) are two faces of the same fact: multiplicity invariance equals count invisibility. The four instances are direct: in each case, the Z -sector itself is the object (or carries the object as Z -internal mechanism).

§7.3 Why W2 is NOT a 5th SDRP instance

It is structurally tempting to register W2 as a 5th SDRP instance, with the V_4 Weil functional as the self-map and Y -sector as the object. This temptation must be rejected. The Y -sector polyhedron is the dodecahedron-icosahedron pair (ZS-F2), which is NOT self-dual: dodecahedron and icosahedron are distinct objects in the dual pair, not a single self-dual object. The SDRP object-self-duality condition fails at the Y -sector layer.

What does hold for W2 is the Cross-Coupling structure of Theorem M31.1, the bilinear form $W_{XYZ}(g)$ of Definition M31.0c, and the Z_2 -parity selection rule of Theorem M31.4: the X - Y - Z cross-coupling in $W_K(g)$ involves the Z -mediator's $\rho_Z = 0$ self-duality (which IS an SDRP-type structure, instance I and II of Table 7.1) and the $J_Z^2 = I$ involution (instance III), but only as components of the cross-coupled expression, not as the full object. W2 is therefore in a syntactic family adjacent to SDRP but distinct: it is a Cross-Coupled expression that incorporates SDRP structure on its Z -mediator component (via $\Pi_Z = (1/2)(I + J_Z)$ using $J_Z^2 = I$), while the full expression is not itself an SDRP instance.

This distinction matters: SDRP closure is internal to Z -Spin (corpus PROVEN four instances), while W2 closure is external (Connes–Burnol–CCM import, corpus PROVEN OPEN). Conflating the two would falsely suggest that W2 closure is internally accessible. Theorem M31.4 Z_2 -parity selection rule is consistent with this: the $J_Z^2 = I$ self-replication of instance III is internal, but it constrains rather than closes the W2 NEG signal — it forces the signal into the $J_Z\text{-EVEN} \times V_4$ -character subspace where external archimedean import is required.

§8. Why Z -Spin Cannot Prove RH Internally

§8.1 NC-M23.1 structural re-derivation

NC-M23.1 (PROVEN, ZS-M23): " Z -Spin does not, and structurally cannot, prove the Riemann Hypothesis from internal data alone." This non-claim has been registered in corpus as a meta-statement following 47+ rounds of cumulative RH exploration. The Cross-Coupling Theorem of ZS-M2 §5 (PROVEN), sharpened by the bilinear-form structure of Definition M31.0c and the Z_2 -parity selection rule of Theorem M31.4, provides a structural re-derivation:

Re-derivation.

Every closed expression in Z -Spin that satisfies the Cross-Coupling Theorem (ZS-M2 §5 PROVEN) is a force formula or its structural homologue. The complete enumeration over six interaction regimes covers all forces. The Riemann Hypothesis is not a force statement; it is a Weil-positivity statement about $\zeta(s)$

zeros on $\sigma = 1/2$. The Weil functional $W_K(g) = B_K(g) - P_K(g)$ does satisfy the Cross-Coupling pattern (Theorem M31.1, this paper), and admits the bilinear-form realization $W_{XYZ}(g) = \langle g_X, (\Pi_Z \otimes I)(B_Y - P_Y)(\Pi_Z \otimes I) g_X \rangle$ (Definition M31.0c) sandwiched by the J_Z -EVEN projector (Theorem M31.4). However, its positivity (the W2 statement) is not forced by Cross-Coupling alone: the J_Z -EVEN sandwich kills the register-level Berry-Keating mechanism (Corollary M31.4a), forcing the W2 closure to flow through the H_{arith} factor where V_4 -character data live. Closure on H_{arith} requires the archimedean place input external to Z-Spin sector decomposition.

§8.2 What this gives the corpus

Theorem M31.1 with the Mathematical Core makes precise where Z-Spin internal structure ends and external mathematics begins. Z-Spin internal structure: the Cross-Coupling Theorem closes all force formulas (ZS-M2 §5 PROVEN, six regimes); the bilinear-form structure $W_{XYZ}(g)$ realizes $W_K(g)$ as a Cross-Coupled object on the Z-mediator projection (Definition M31.0c); the Z_2 -parity selection rule constrains the operator content (Theorem M31.4). External mathematics required: archimedean / Lie-theoretic / spectral imports for the three RH walls. The boundary between internal and external is precisely the Cross-Coupling closure boundary, refined by the J_Z parity sector.

This is consistent with ZS-M23 §4 (Three Z-Spin Contributions to RH, PROVEN): C1 (i-tetration anti-symmetric phase), C2 (Wilson conjugate pair), C3 ($j = 1/2$ spinor 4π closure) are corpus PROVEN at the operator/algebraic level, but they do not by themselves close any RH wall. The walls require external imports.

§9. Falsification Gates

Ten falsification gates are pre-registered. Each specifies a falsifying condition, the consequence within ZS-M31, and the verification status.

Table 9.1. ZS-M31 falsification gates (5 syntactic + 3 v1.1 computational + 2 Edge analytic).

Gate	Falsifying condition	Consequence	Status
F-M31.1 (theoretical)	Cross-Coupling Theorem (ZS-M2 §5) is shown to fail for any of six regimes	Theorem M31.1 falsified	PASS (ZS-M2 PROVEN)
F-M31.2 (numerical)	Independent recomputation of ZS-M26 12-grid yields different sign at $>10^{-3}$	Theorem M31.2 falsified	PASS (ZS-M26 PROVEN)
F-M31.3 (structural)	Internal Z-Spin closure of W2 without external archimedean import	Theorem M31.3 weakened	PASS (ADS-5/6/7)
F-M31.4 (interpretive)	5th SDRP instance with W2 as the object	SDRP scope extended	PASS (Y-sector non-self-dual)
F-M31.5 (anti-overclaim)	New free parameter or new physical prediction beyond LOCKED inputs	Zero-FP discipline violated	PASS (Table 2.1 audit)
F-M31.6 (mathematical, v1.1)	$\Pi_Z = (1/2)(I + J_Z)$ shown not idempotent under $J_Z^2 = I$	Definition M31.0c invalid	PASS (B-1 algebraic)

F-M31.7 (simulation, v1.1)	Any of 18 (a, t1, t2) tests yields across-channel variance < 0.1	Lemma M31.0 falsified	PASS (NS-1: 18/18, max var 13.011)
F-M31.8 (numerical, v1.1)	Any (a, t) grid point fails $n^* = 1/2$ classification	Corollary M31.2a falsified	PASS (CT-1: 12/12)
F-M31.9 (algebraic, Edge)	$\prod_Z K^{\{-J\}} \prod_Z \neq 0$ for some J_Z -ODD K under $J_Z^2 = I$	Theorem M31.4 falsified	PASS (E1: 50 random Hermitian tests)
F-M31.10 (analytic, Edge)	$n^*_{RL} = 2\sqrt{(\ln 2)/\pi} \notin (0.4502, 0.7118)$	Lemma M31.2b falsified; threshold not corpus-natural	PASS (E5: $0.5300 \in$ interval)

§10. Verification Suite (36/36 PASS)

Verification combines 23 inheritance tests (algebraic identities at machine precision and 50-digit mpmath inheritance from upstream) plus 8 v1.1 computational tests plus 5 Edge analytic tests executed in `zs_m31_verify.py` (companion script). The companion script is publicly available at https://github.com/KennyKang-git/zspin/tree/main/verify_scripts. Reproduction: `python3 zs_m31_verify.py`. Expected output: 36/36 PASS, exit code 0. Approximate runtime: < 5 seconds. No external dependencies beyond Python 3.10+ standard library plus numpy (for the 11×11 register tests of category E, J_Z parity decomposition).

Table 10.1. ZS-M31 verification suite (36/36 PASS).

Cat.	Tests	Scope	Type
A: Locked Inputs	5/5 PASS	$A = 35/437$; $(Z, X, Y) = (2, 3, 6)$; $Q = 11$; $\rho_Z = 0$; $L_{XY} \equiv 0$; $J_Z^2 = I$	Inheritance
B: Bilinear Form	3/3 PASS	$\prod_Z^2 = \prod_Z$ (B-1); $(B_Y - P_Y)$ Hermiticity (B-2); Plancherel $\hat{g}_{\{a,t\}}(\xi)$ (B-3)	New computational
C: Cross-Coupling	3/3 PASS	$\sin^2 \theta_W$ three-in-one; α_s three-in-one; α_2 three-in-one	Inheritance
D: $W_K(g)$ Decomposition	4/4 PASS	B_K archimedean; P_K prime; V_4 orthogonality; Robin family bracket	Inheritance
E: Z_2 -Parity Selection (Edge)	3/3 PASS	$\prod_Z K^{\{+J\}} \prod_Z = \prod_Z K \prod_Z$ (E1); $\prod_Z K^{\{-J\}} \prod_Z = 0$ (E2); $\sigma_x^Z J_Z$ -ODD (E3)	New analytic
E': 5/12 NEG Localization	3/3 PASS	Probe W2 grid PROVEN; small-(a, t) NEG locus; large-t averaging PSD	Inheritance
CT: Cycles Threshold	2/2 PASS	$n^* = 1/2$ classifies 12/12 (CT-1); separator interval $\ni 1/2$ (CT-2)	New computational
CT': Analytic Threshold (Edge)	2/2 PASS	$n^*_{RL} = 2\sqrt{(\ln 2)/\pi} \approx 0.5300$ (E4); $0.5300 \in (0.4502, 0.7118)$ (E5)	New analytic
F: Three-Wall Homogeneity	3/3 PASS	W1 Frame Layer; W2 Sector Layer; W3 Protocol Layer	Inheritance
G: SDRP	4/4 PASS	ZS-F9 $V \leftrightarrow F$; ZS-F12 TDO; ZS-A9 BT-engine; ZS-F13 Möbius	Inheritance
NS: Non-Separability	3/3 PASS	18/18 sum-form falsification (NS-1); max variance 13.011 (NS-2); sample 4.526 (NS-3)	New computational
H: Anti-Numerology	1/1 PASS	Zero new free parameters audit (Table 2.1)	Inheritance
TOTAL	36/36 PASS	Machine precision + 50-digit mpmath inheritance + 8 computational + 5 Edge analytic	23 + 8 + 5

§11. Conclusion

This paper establishes the Cross-Coupling reading of the V_4 Weil functional Wall W2 as the corpus-consistent reformulation that recovers the Three-in-One pattern of every Z-Spin force formula. Four principal results are established.

Theorem M31.1 (W2 Three-in-One Decomposition, DERIVED) places $W_K(g) = B_K(g) - P_K(g)$ explicitly in the Cross-Coupling pattern, with X (test function), Y (V_4 characters and conductors), and Z (Robin family invariance via $\rho_Z = 0$) appearing simultaneously. Definition M31.0c makes this quantitative: the bilinear form $W_{XYZ}(g) = \langle g_X, (\Pi_Z \otimes I)(B_Y - P_Y)(\Pi_Z \otimes I) g_X \rangle$ realizes $W_K(g)$ as a Cross-Coupled operator on the Z -mediator projection $\Pi_Z = (1/2)(I + J_Z)$ (corpus PROVEN $J_Z^2 = I$).

Theorem M31.2 (5/12 NEG Localization, DERIVED) traces the corpus PROVEN sign distribution to the small- (a, t) region. Corollary M31.2a (Cycles Threshold, DERIVED) sharpens this to $n^* = 1/2$ separating NEG and PSD with 12/12 accuracy; Lemma M31.2b (Analytic Threshold, DERIVED) confirms this from Plancherel half-decay: $n^*_{RL} = 2\sqrt{(\ln 2)/\pi} \approx 0.5300 \in (0.4502, 0.7118)$, with the corpus-natural $1/2$ and the analytic 0.5300 both classifying 12/12 — the $1/2$ choice is therefore the corpus-rounded form of an analytically determined threshold, not a numerological match.

Theorem M31.3 (Three-Wall Cross-Coupling Homogeneity, DERIVED) demonstrates that W_1, W_2, W_3 are three Layer-specific instances of the same Cross-Coupled syntax, with W_3 already closed via Kostant cubic Dirac (ZS-M27) and W_1, W_2 awaiting analogous external archimedean imports.

Theorem M31.4 (Z_2 -Parity Selection Rule, DERIVED), the Edge result of v1.0, is a direct consequence of corpus PROVEN $J_Z^2 = I$: the bilinear form $W_{XYZ}(g)$ sees only the J_Z -EVEN component of $(B_Y - P_Y)$ on the register factor. Combined with ZS-F0 Corollary 8.16 PROVEN (σ_x^Z is J_Z -ODD), Corollary M31.4a forces the W_2 NEG signal to flow through the J_Z -EVEN $\times V_4$ -character subspace of $H_{BFV} \otimes H_{arith}$ — the Berry-Keating commutator on the register factor is invisible to the bilinear form. This sharpens the no-go content of ADS-5/6/7 PROVEN: no internal escape route on the register factor can produce the W_2 NEG signal; closure is forced through external archimedean mathematics on H_{arith} .

Lemma M31.0 (Non-Separability, DERIVED, 18/18 PASS) empirically falsifies any sum-form decomposition $W_K(g_{\{a,t\}}) = F_X(a, t) + F_Y(\chi, q_\chi, a_\chi) + F_Z(\rho_Z, J_Z)$ on the ZS-M26 §5.3 PROVEN data, with maximum across-channel variance $13.011 \gg$ measurement noise (≈ 0.05). The bilinear-form structure plus Lemma M31.0 plus Theorem M31.4 jointly elevate the syntactic Cross-Coupling reading to a quantitative result with PROVEN operator structure, DERIVED separability falsification, and DERIVED Z_2 -parity selection rule.

Five corollaries follow. C1 registers the Self-Dual Replication Principle (SDRP) as the unification of four corpus instances. C2 clarifies that W_2 is NOT a 5th SDRP instance. C3 re-reads the ζ -pole and Burnol corrections as facets of one external archimedean closure path. C4 places W_3 closure and W_2 OPEN status on the same structural footing. C5 structurally re-derives NC-M23.1.

This paper introduces zero new free parameters and zero new physical predictions. The contribution is the structural reformulation of W2 under Cross-Coupling discipline plus the Mathematical Core (bilinear form, Non-Separability Lemma, cycles threshold) plus the Edge results (Z_2 -parity selection rule with Berry-Keating invisibility corollary; Riemann-Lebesgue analytic threshold confirming the corpus-natural $1/2$). Ten falsification gates F-M31.1 through F-M31.10 are pre-registered; all PASS. Verification 36/36 PASS (23 inheritance + 8 computational + 5 Edge analytic). NC-M23.1 is preserved verbatim and structurally deepened: the Cross-Coupling boundary, refined by the J_Z parity sector, is the precise locus where Z-Spin internal structure ends and external mathematics begins.

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude) for mathematical verification, structural cross-referencing, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions.

The verification script `zs_m31_verify.py` is publicly available at https://github.com/KennyKang-git/zspin/tree/main/verify_scripts. Reproduction: `python3 zs_m31_verify.py`. Expected output: 36/36 PASS, exit code 0. Approximate runtime: < 5 seconds. Dependencies: Python 3.10+ standard library plus numpy (Edge category E uses numpy for 11×11 register parity decomposition tests).

Inheritance tests inherit verification from upstream corpus papers: Cross-Coupling Theorem (ZS-M2 §9, 26/26 PASS), $W_K(g)$ sign distribution diagnostic (ZS-M26 Probe W2, PROVEN), W3 Kostant Dirac closure verification (ZS-M27, 24/24 PASS), V_4 Schur orthogonality (ZS-M26, 24/24 PASS), and the SDRP four-instance verification (ZS-F9, ZS-F12, ZS-A9, ZS-F13).

Appendix A. The Six Cross-Coupling Regimes

ZS-M2 §5 (PROVEN by enumeration) establishes the Cross-Coupling Theorem over six interaction regimes. Each regime exhibits a single closed expression in which X, Y, Z appear simultaneously.

Table A.1. Six Cross-Coupling regimes (reproduced from ZS-M2 §5, PROVEN).

Force	X component	Z component	Y component
α_s	$Q = Z \cdot X + X + Z$ (indirect)	Z in Q ($\beta_0(Z) = 1$)	$(V+F)_Y + 1 = 93$
$\sin^2\theta_W$	$V_X = 24$ ($ O_h = 48$)	x^* (Berry phase)	$(V+E+F)_Y/2 = 91$
α_2	$(V+F)_X = 38$	$S = Z \cdot (E_Y/E_X)$	$Y = 6$
G_{eff}	δ_X in A	Z in factorization	δ_Y in A
DM (a_0)	halo (X-gradient)	c, H_0 (Z-mediated)	Y in $a_0 = cH_0/Y$
DE (p_Λ)	implicit via A	Z in A	Y-attractor ε^*

Each row exhibits a closed expression in which all three sectors contribute simultaneously. ZS-M31 Theorem M31.1 places $W_K(g)$ on the same structural footing; Definition M31.0c realizes the closed expression as the bilinear form $W_{XYZ}(g)$; Theorem M31.4 selects the J_Z-EVEN sector.

Appendix B. Why the Two-Mechanism Reading Fails

The two-mechanism reading separates W2 NEG into (a) a ζ -pole channel handled by archimedean pole correction, and (b) a finite-prime conductor channel handled by external Burnol correction. This reading is structurally tempting because the corpus does identify two distinct external mathematical traditions for the two corrections.

However, the reading violates Cross-Coupling Theorem (ZS-M2 §5 PROVEN). The $W_K(g)$ expression is a single closed object: it is not a sum $W_K = W_K^{\{\text{arch}\}} + W_K^{\{\text{prime}\}}$ of independent mechanisms, but a single difference $B_K(g) - P_K(g)$ where B_K and P_K interact non-trivially through the test function g . Separating the two corrections as independent mechanisms treats the cross-coupling as decomposable, which contradicts the Cross-Coupling structure. Lemma M31.0 makes this quantitative: the empirical 18/18 falsification of any sum-form decomposition prevents Reading B at the data level.

ZS-M28 §7.2 (cited) and ZS-M22 §6.6.5(a) (PROVEN diagnostic) jointly establish that pole correction and conductor correction are two facets of one external archimedean closure path: the Connes 2000 archimedean place, sharpened by Burnol 2002, 2004, finalized by Connes–Consani–Moscovici 2024. The three external references constitute one closure tradition.

Theorem M31.1 of this paper (DERIVED) recovers Cross-Coupling consistency by reading $W_K(g)$ as one expression and the external import as one closure. Definition M31.0c realizes this as the bilinear form $W_{XYZ}(g)$, Theorem M31.4 selects the J_Z -EVEN content, and Corollary M31.4a localizes the W2 NEG signal to the H_{arith} factor. The two-mechanism reading is rejected as a Cross-Coupling violation; the one-mechanism Y-incompleteness reading is rejected as an SDRP-instance miscategorization (cf. §7.3).

Appendix C. Non-Separability Lemma — Full 18-Test Table

Complete computational record for Lemma M31.0. Each row is one $(a, t1, t2)$ test with $t1 < t2$; $\Delta_\chi = W_\chi(g_{\{a,t1\}}) - W_\chi(g_{\{a,t2\}})$ is computed for each of the 4 V_4 channels (ζ no-pole, $\chi_{\{-3\}}$, $\chi_{\{-11\}}$, $\chi_{\{33\}}$). The variance $\max(\Delta_\chi) - \min(\Delta_\chi)$ measures across-channel inconsistency under the sum-separability hypothesis. Empirical floor for measurement noise: ≈ 0.05 (P_{max} stability per ZS-M26 §5.3 PROVEN). Falsification threshold: > 0.1 (twice the noise floor).

Table C.1. Full Non-Separability test (18/18 violations). Source: ZS-M26 §5.3 Table 5.2 PROVEN.

a	t1	t2	Δ_χ for ($\zeta, \chi_{\{-3\}}, \chi_{\{-11\}}, \chi_{\{33\}}$)	max-min
0.2	0	1	-8.641 / +0.310 / +0.054 / +1.187	9.828
0.2	0	5	-7.428 / +1.802 / +1.735 / +0.234	9.230
0.2	0	14.13	-10.341 / +2.670 / +0.759 / +0.350	13.011
0.2	1	5	+1.213 / +1.492 / +1.681 / -0.953	2.634
0.2	1	14.13	-1.700 / +2.360 / +0.705 / -0.837	4.060
0.2	5	14.13	-2.913 / +0.868 / -0.976 / +0.116	3.781
0.5	0	1	-2.067 / +0.209 / -0.406 / +0.298	2.365
0.5	0	5	-2.934 / +1.592 / +1.250 / -0.741	4.526
0.5	0	14.13	-4.381 / +1.911 / -0.069 / -0.565	6.292
0.5	1	5	-0.867 / +1.383 / +1.656 / -1.039	2.695
0.5	1	14.13	-2.314 / +1.702 / +0.337 / -0.863	4.016
0.5	5	14.13	-1.447 / +0.319 / -1.319 / +0.176	1.766

1.0	0	1	-0.619 / +0.138 / -0.211 / -0.071	0.757
1.0	0	5	-1.526 / +1.235 / +0.990 / -1.018	2.761
1.0	0	14.13	-2.215 / +1.291 / +0.275 / -0.962	3.506
1.0	1	5	-0.907 / +1.097 / +1.201 / -0.947	2.148
1.0	1	14.13	-1.596 / +1.153 / +0.486 / -0.891	2.749
1.0	5	14.13	-0.689 / +0.056 / -0.715 / +0.056	0.771

All 18 rows show variance > 0.1 , with maximum variance 13.011 at $(a=0.2, t_1=0, t_2=14.13)$. The minimum variance is 0.757 at $(a=1.0, t_1=0, t_2=1)$, still $15\times$ above the 0.05 noise floor and $7.6\times$ above the 0.1 falsification threshold. The sum-separability hypothesis is rejected at 18/18 tests with high statistical confidence. [Verified: NS-1, NS-2, NS-3 in `zs_m31_verify.py`.]

Appendix D. Edge Theorem M31.4 — Explicit Z_2 -Parity Decomposition Verification

Numerical verification of Theorem M31.4 on 50 random 11×11 Hermitian operators $K = K^\dagger$ chosen with seed 42. For each K we compute $K^{\wedge\{+_J\}} = (1/2)(K + J_Z K J_Z)$, $K^{\wedge\{-_J\}} = (1/2)(K - J_Z K J_Z)$, and verify the three identities of the Theorem at machine precision (Frobenius norm tolerance $1e-12$):

Identity	Frobenius norm of LHS – RHS	Status
$K^{\wedge\{+_J\}} + K^{\wedge\{-_J\}} = K$	$< 1e-15$ (50/50)	PASS
$\Pi_Z K \Pi_Z = \Pi_Z K^{\wedge\{+_J\}} \Pi_Z$	$< 1e-14$ (50/50)	PASS
$\Pi_Z K^{\wedge\{-_J\}} \Pi_Z = 0$	$< 1e-15$ (50/50)	PASS
$J_Z \sigma_x^{\wedge Z} J_Z = -\sigma_x^{\wedge Z}$ (Cor 8.16)	0 exactly	PASS
$\Pi_Z \sigma_x^{\wedge Z} \Pi_Z = 0$ (Cor M31.4a)	0 exactly	PASS

All five identities PASS at machine precision on all 50 random tests. The result is algebraic (independent of the specific K) and derives only from corpus PROVEN $J_Z^2 = I$ (ZS-F0 §8.6 Theorem 8.13). Status: DERIVED. The verification is reproducible from `zs_m31_verify.py` Edge category E (E1-E3) plus the Plancherel/threshold tests E4-E5.

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Internal corpus references follow Z-Spin convention: ZS-[theme code] [paper number]; status tags in parentheses where relevant. External references follow APS numerical-citation order.

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Version History

v1.0 (March 2026): Initial public release. Four principal theorems M31.1, M31.2, M31.3, M31.4 established under Cross-Coupling Theorem discipline (ZS-M2 §5 PROVEN). Mathematical Core (§4.0) introduces three explicit objects: per-channel V_4 decomposition (Definition M31.0a, inheriting ZS-M22 §6.6.5(b) PROVEN), test function and Plancherel transform (Definition M31.0b, DERIVED from standard Fourier identity), bilinear form $W_{XYZ}(g)$ on $H_{BFV} \otimes H_{arith}$ with $\Pi_Z = (1/2)(I + J_Z)$ (Definition M31.0c, DERIVED from ZS-F0 §8.6 PROVEN $J_Z^2 = I$). Lemma M31.0 (Non-Separability, DERIVED, 18/18 PASS) empirically falsifies any sum-form decomposition with maximum across-channel variance $13.011 \gg$ measurement noise 0.05. Corollary M31.2a (Cycles Threshold, DERIVED) gives $n^* = 1/2$ separating 12/12 grid points; Lemma M31.2b (Riemann-Lebesgue Analytic Threshold, DERIVED) derives $n^*_{RL} = 2\sqrt{(\ln 2)/\pi} \approx 0.5300$ from Plancherel half-decay, confirming corpus-natural $1/2$ is not numerological. Theorem M31.4 (Z_2 -Parity Selection Rule, DERIVED) — the Edge result — proves $\Pi_Z K \Pi_Z = \Pi_Z K^{+J} \Pi_Z$ by direct algebraic computation using $J_Z^2 = I$; Corollary M31.4a (Berry-Keating commutator invisibility) follows from ZS-F0 Cor 8.16 PROVEN. Five corollaries C1–C5 registered (SDRP, etc.). Self-Dual Replication Principle (SDRP) registered as the unification of four corpus instances (ZS-F9 §3, ZS-F12, ZS-A9, ZS-F13). Verification 36/36 PASS (23 inheritance + 8 computational + 5 Edge analytic). Companion script `zs_m31_verify.py` provides reproducible PASS at machine precision. Ten falsification gates F-M31.1 through F-M31.10 pre-registered (5 syntactic + 3 v1.1 computational + 2 Edge analytic). Zero new free parameters; zero new physical predictions. $A = 35/437$ and $Q = 11$ remain the sole geometric inputs. The contribution is the structural reformulation of W2 under Cross-Coupling discipline plus the Mathematical Core (bilinear form, Non-Separability Lemma, cycles threshold) plus the Edge results (Z_2 -parity selection rule with Berry-Keating invisibility; Riemann-Lebesgue analytic threshold). NC-M23.1 (Z-Spin cannot prove RH from internal data alone) preserved verbatim and structurally deepened: the Cross-Coupling boundary refined by the J_Z parity sector is the precise locus where Z-Spin internal structure ends and external mathematics begins. (Consolidated from internal Z-Spin Collaboration research notes through cumulative exploration on W2 structural reading and Mathematical Core derivation, March 2026.)